

```
In [16]: # Import Libraries
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
```

```
In [18]: import pandas as pd

data = {
    'Month': [1,2,3,4,5,6,7,8,9,10],
    'Sales': [1000,1200,1400,1500,1700,1800,2000,2200,2400,2600]
}

df = pd.DataFrame(data)

print(df)
```

	Month	Sales
0	1	1000
1	2	1200
2	3	1400
3	4	1500
4	5	1700
5	6	1800
6	7	2000
7	8	2200
8	9	2400
9	10	2600

```
In [20]: print("First 5 Records:")
print(df.head())
```

First 5 Records:

	Month	Sales
0	1	1000
1	2	1200
2	3	1400
3	4	1500
4	5	1700

```
In [21]: X = df[['Month']]
y = df['Sales']
```

```
In [22]: model = LinearRegression()
```

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In [23]: model.fit(X, y)
```

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Out[23]: ▼ LinearRegression ⓘ ?
          ► Parameters
```

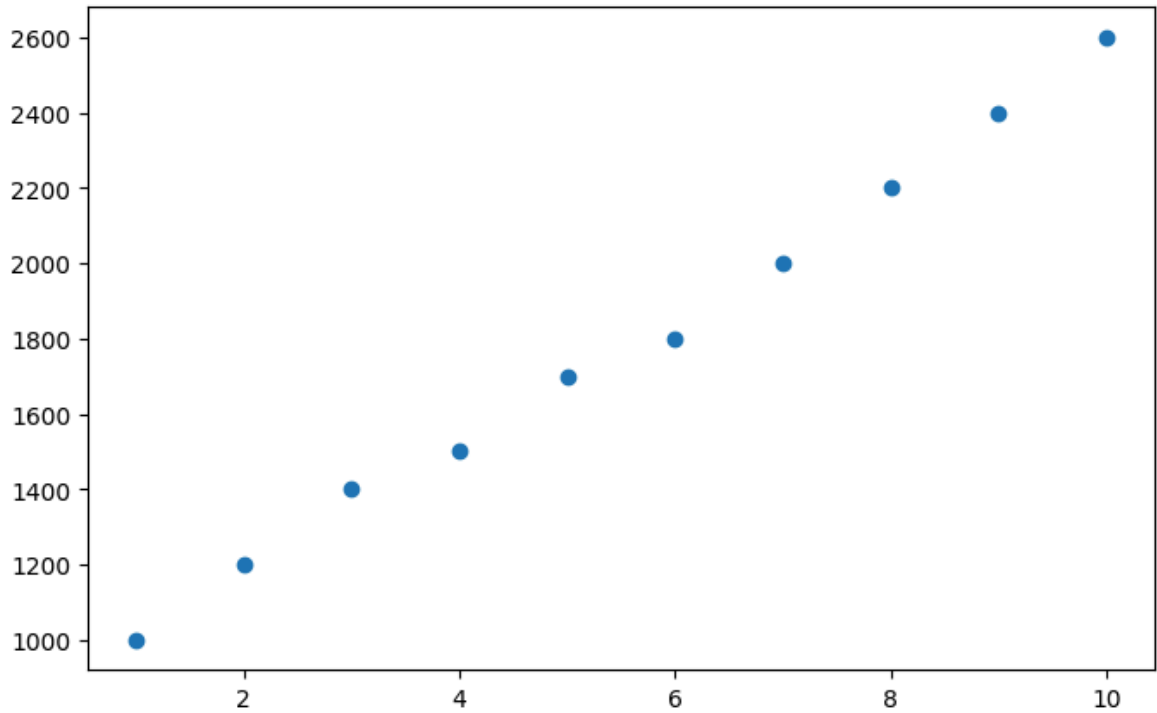
```
In [24]: future_months = pd.DataFrame({'Month': [11, 12, 13, 14, 15]})
predictions = model.predict(future_months)
```

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In [25]: print("\nSales Forecast:")
for month, sales in zip(future_months['Month'], predictions):
    print(f"Month {month}: {sales:.2f}")
```

Sales Forecast:
Month 11: 2726.67
Month 12: 2898.79
Month 13: 3070.91
Month 14: 3243.03
Month 15: 3415.15

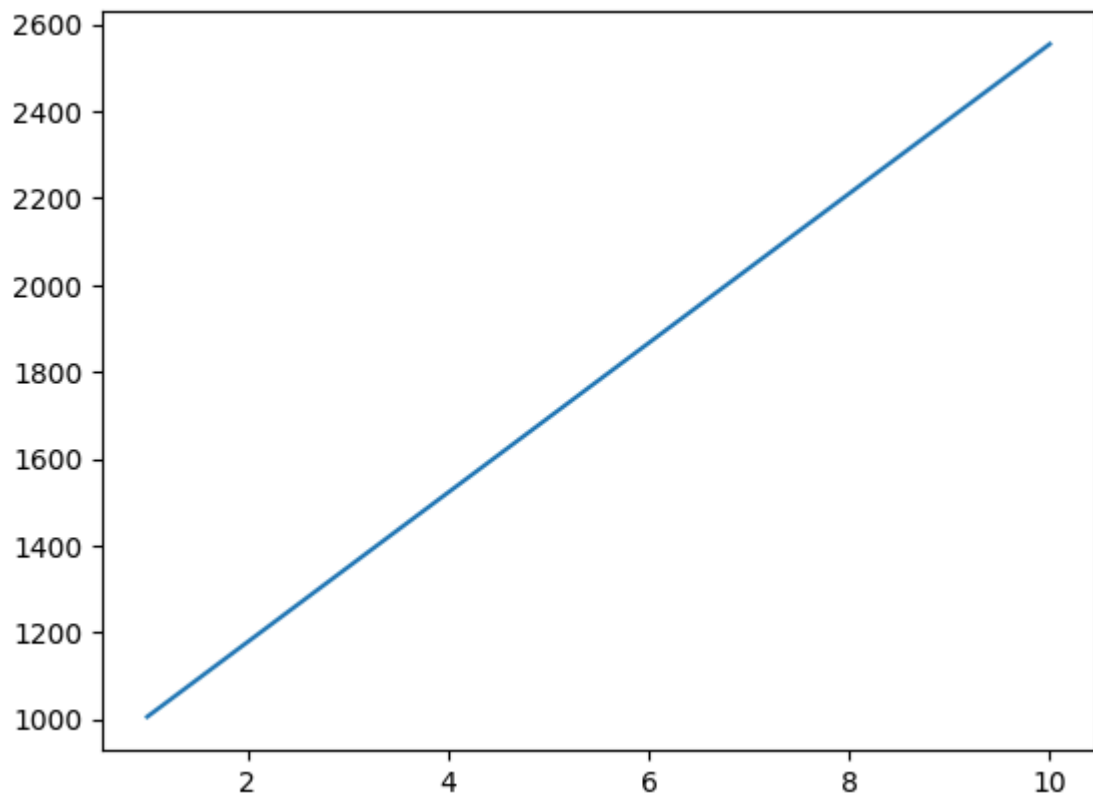
```
In [26]: plt.figure(figsize=(8, 5))  
plt.scatter(X, y, label='Actual Sales')
```

Out[26]: <matplotlib.collections.PathCollection at 0x2c24d522120>

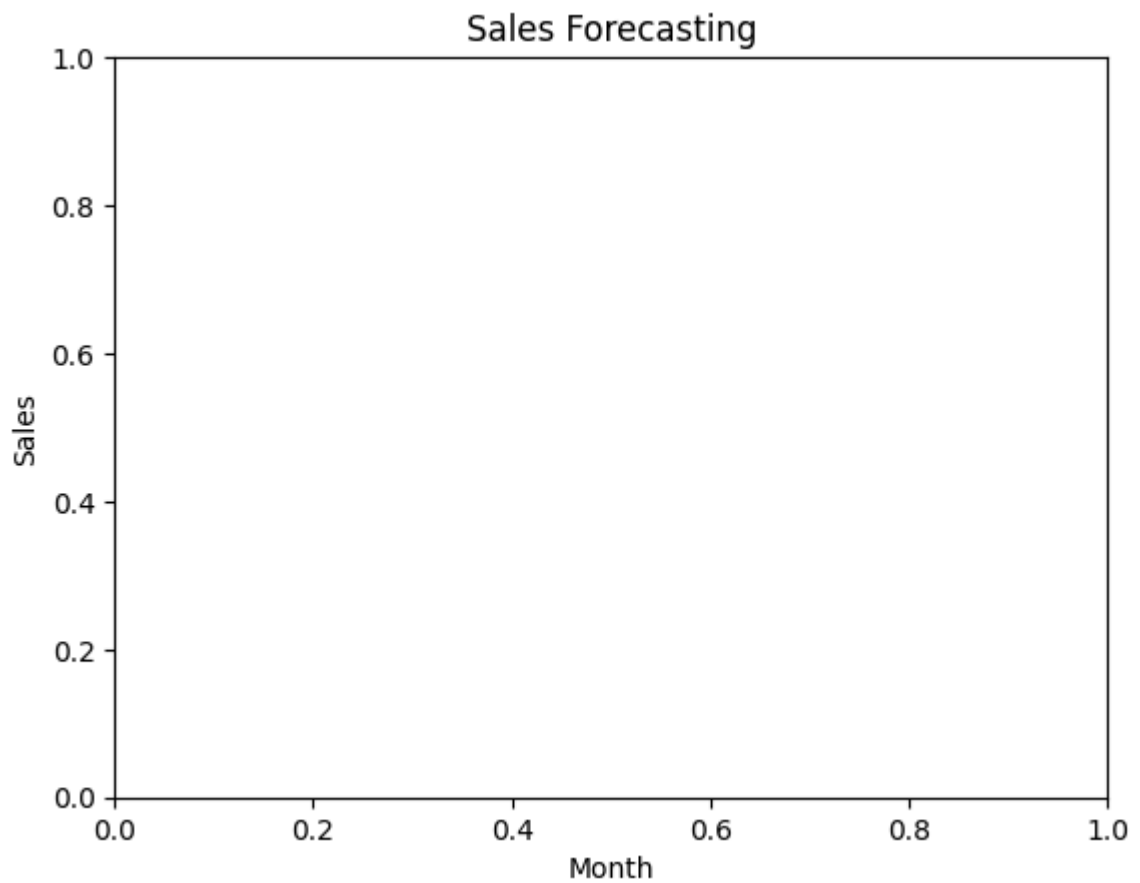


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In [27]: plt.plot(X, model.predict(X), label='Regression Line')
```

Out[27]: [<matplotlib.lines.Line2D at 0x2c24d7c0050>]



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In [29]: plt.xlabel("Month")
plt.ylabel("Sales")
plt.title("Sales Forecasting")
plt.show()
```



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In [ ]:
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